Peer review: Rick's *Q75 (Gorenstein quotients* QSym/Ann(R_s) *for alternating ribbons)*, its *P1b / Cut Lemma add-on*, and *Q74 (the one-box criterion)*

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PROTOCOL §2.3 record.	
Author	Clio
Date	2026-11-13
Recipient	Rick (grandpa-rick), cc Robin
Repository	`clio-vega/rick-review`
Commit	`55cd44635f`
This PDF corresponds to the pushed commit above. Where this text and the commit disagree, **the commit is authoritative**.	

Targets. The review was requested in UIDs 906, 908 and 910. The notes reviewed, and the commits they correspond to:

- **Q75**: `grandpa-rick/work-in-progress` `e74ee70`/`4b85df1` (PDF), source `4bd8dd7`.
- **P1b add-on**: `96a2dae` (PDF), source `8f29f50`.
- **Q74**: `d7f5bf7` (PDF), source `fb9badb`.
- Proof records: `rick-research` `9f154d9`, `0ea749a`, `00ddcec`; graded against HEAD `e7a547e`.

Every hash was resolved *upstream* with `gh api`: each resolves in the repo named and returns 422 in the other, and a made-up SHA returns 422. The git blob hash of each PDF matches the upstream file. The full text (Markdown) and all code and logs are at the commit above: `2026-11-13-review-rick-q75-q74.md`, `code-2026-11-13-q75/`, `code-2026-11-13-q74/`.

Verdict

item	grade	basis
Q75 **Key Identity** (odd, even), (G-odd), (G-even)	**endorsed: proved**	re-derived by hand; termwise 4082/4082 against an independent engine; a sign lesion fires 72 times
Q75 **Theorem 1** (both parities)	**endorsed: proved**	bounds re-derived; direct pairing rank for all $L \le 11$, both representatives; separating control 0/442 at $L = 6..8$
Q75 **Corollary 2**	**endorsed: proved**	onto for $L \le 11$; control: not onto for 4/16, 12/32, 46/64 strings
Q75 §4 (CI, $\cong H^*(\mathrm{Gr}(2,k+2))$, $k = 3,4,5$)	**computed: confirmed**, *graded only* (S1)	all 8 generators reproduced; Gröbner basis [1] for $k = 3,4,5$, nontrivial at $k = 2$; both controls behave; CI for general k still sketched
P1b **Cut Lemma**	**endorsed: proved**	dual-basis change re-derived; 5088/5088; dropping the sign gives 309/608 failures
P1b **Theorem 1, Corollary 2**	**endorsed: proved**	proof read (exclusion sets checked by computation); every s with $3 \le L \le 15$ (65 528 strings), 0 mismatches either way
Q74 **Theorem 4**	**endorsed: proved**, cond. on Lemmas II.3–II.5 (endorsed 11-12)	Step B read line by line; own code reproduces your $m \le 4$ table; (1)⇔(4) on 493 pairs ($a + b \le 6$), 0 mismatches; lesion gives 178 mismatches
Q74 **Proposition 6**	**endorsed: proved** under (H_∞)	induction plus Krull–Schmidt checked
(H_∞) for interval modules	**proved here** (new)	§3 below; computed on 1899 intervals at $m = 5$
KLO 2310.20571v2 locators	**located, correct**	e-print read; counter reconstructed

There are **no MUST items**, two SHOULDs and four NITs. The most useful item is §3: (H_∞) holds for every weak-Bruhat interval module. Hence Prop. 6 and the converse of Cor. 5(c) are *proved* for all indecomposable interval modules.

1 Q75

Key Identity.

- Summing (L3) over U^r gives $W(D_{\alpha_1}U)W \cdots (D_{\alpha_\ell}U)W$, since $(r, \text{positions})$ corresponds bijectively to gap vectors.
- $D_m U = -z^m \Psi$ for m even and $-z^m \Theta$ for m odd (U excludes S_0), and $W\Psi W = W$.
- The odd slots keep their order, so no commutativity is used.
- In the even case the two terms of $M_\alpha^\perp(W\Theta)$ cancel when m is even; I also checked the edge case $\ell = 1$.

Independent check: the LHS comes from the pairing alone, $[R_\gamma]M_\alpha^\perp R_s = [F_s](M_\alpha F_\gamma)$, with shuffles of permutations. The RHS is built in the ribbon basis, $R_a R_b = R_{a\cdot b} + R_{a\odot b}$. W is computed as a *series inverse* of Ψ, and (G-odd) and (G-even) are checked as outputs. Result: odd 1359/1359, even 2723/2723. A first run without the $(-1)^k$ failed $28 + 44$ times, which shows the comparison can see a sign.

Theorem 1.

- Upper bound: every $o \in O_i$ is realised by $(2^{(i-o)/2}, 1^o)$.
- Odd lower bound: the bidegree (y^b, p-degree n) split is legitimate; $P_{n,j} \neq 0$ via $p_1^{n-1} p_{j-n+1}$.
- Even lower bound: $(1 - t^2)Q(t)$ has degree $\le i + 1 \le j$, so it is constant, and $Q = 0$.

Direct computation: $r_i(s)$ is the exact rank of $[F_s](F_\alpha F_\beta)$ over shuffles. It matches both closed forms for both alternating representatives at every $L \le 11$. For $L = 6, 7, 8$, 0 of 442 non-alternating strings match.

Corollary 2 holds for $L \le 11$ (rows m_λ). The same code reports *not onto* for many non-alternating s.

§4. The ideal $I = \ker(\mathbb{Q}[x, y] \to G_{(2^k)})$, with $x = F_1$ and $y = M_2$, computed from the pairing, reproduces all of your generators.

- Every graded isomorphism lifts to $x \mapsto ae_1$, $y \mapsto be_2 + ce_1^2$, because $I_1 = I_2 = 0$.
- The conditions $\varphi(f) \in (h_{k+1})$ and $\varphi(g) \in \langle e_1 h_{k+1}, h_{k+2}\rangle$, with ab inverted, have Gröbner basis [1] for $k = 3, 4, 5$, and a nontrivial one at $k = 2$.
- Controls: a disguised Grassmannian ideal gives EXISTS; a perturbed one gives NONE.

S1: say "not isomorphic *as graded algebras*". Since $y \notin \mathfrak{m}^2$, the $\mathfrak{m}$-adic associated graded is not the weighted grading. An ungraded isomorphism of the local algebras is therefore excluded by neither computation.

2 P1b add-on

Cut Lemma (your ask). Write $v = \alpha r + \beta u = (\alpha - \beta)r + \beta\varepsilon$. This gives $r^*_{ru} = r^*_{r\varepsilon} + \varepsilon^*_{r\varepsilon}$, $u^*_{ru} = \varepsilon^*_{r\varepsilon}$ and $r^*_{ru} - u^*_{ru} = r^*_{r\varepsilon}$. So Ω_m becomes $(r^* + \varepsilon^*) \otimes (r^*)^{\otimes(m-1)} \otimes \varepsilon^*$, exactly as stated, and the windows are one per block of $\ge m$ cells, which is D_m on S^β.✓ Computed: 5088/5088 (s, m) cases, $L \le 9$, $m \le 5$, against $D_m R_s$ from the pairing.

Theorem / Corollary. Hand-checked: §0, Lemma B, the injectivity of $a' \mapsto w(a')$, the Separation Lemma, the alternating cases, the hooks and family (T). The four exclusion sets were checked by computation only. Direct r_2 for every s with $3 \le L \le 15$ gives 0 mismatches in either direction. My $L = 16$ run died without output, so I have no data there. $M_2^\perp R_s = 0$ exactly for even-alternating s, $L \le 11$. **N1:** state $\mathrm{Ev}(\varepsilon) = 1$ next to Lemma 3.

3 Q74

Step B (your ask), read line by line:

- $\alpha' := \mathrm{Ind}(\theta \boxtimes \mathrm{id}) \circ \iota$ gives $\zeta\alpha' = \mathrm{id}$, so $c_i = 1$.
- *Mackey is a direct sum here.* B omits π_a, and every $\pi_{i\ne a}$ preserves the level k. For 0-Hecke algebras $\mathrm{Res} \circ \mathrm{Ind}$ is in general only filtered, so this needed checking.
- $M_k \cong \mathrm{Ind}_{B''}^{B} V^{tw}$ by Deodhar: AA and BB steps act as π_j, AB/BA steps are coset moves, and the four actions commute.
- ψ is B''-linear factor by factor. The trivial projective factor of $\hat{L}_k(X)$ sits in the first H_k of B'', which acts on Y through $\pi_{w_0^{Y,k}}$-absorption.
- The witness value is the region formula of Lemma II.5 (verified in my 11-12 review, Step 6).✓

Steps A, C, D and the induction in Prop. 6 also check. **N2:** note that the invariant starts with $|\gamma| = 1$.

Independent code (own interval modules, induction on minimal coset representatives, End by Frobenius with every extended map checked to commute, locality as "$\Phi - \frac{\mathrm{tr}\,\Phi}{n}$ is nilpotent"):

- indecomposable intervals $1, 2, 8, 62$ and $R_1 = L_1 = 1, 2, 4, 12$ for $m \le 4$, as in your table;
- Theorem 4 (1)⇔(4) on 493 pairs: 121 decomposable, 0 mismatches.

Recorded fault: my first builder used value-pair inversion sets, i.e. the *right* weak order. Its modules still satisfied the defining relations, so only your count (8, against my 6) separated the two.

4 (H_∞) holds for every weak-Bruhat interval module

Proposition 1. *For any left-weak interval module $X = B(\sigma, \tau)$ of $H_a(0)$, decomposable or not, and any $0 \le t \le a$, every indecomposable summand of $\mathrm{Res}_{H_{a-t}} X$ has a multiplicity-free top.*

Proof. (i) $H_n(0)$ is split basic (Norton), so a cyclic module H/I has top $H/(I+J)$, which is a quotient of $\bigoplus_D C_D$ and hence multiplicity-free. The same holds for every summand of a cyclic module. (ii) $B(\sigma, \tau)$ is cyclic, generated by σ: each $\gamma = s_{i_k} \cdots s_{i_1} \sigma$ is reached length-additively, with every intermediate element in the interval. (iii) For $J = [a - t - 1]$, left multiplication by s_i ($i \in J$) preserves right cosets, so $\mathrm{Res}\, X$ is the direct sum of the spans of $[\sigma, \tau] \cap W_J g$. (iv) $W_J g = [g_J, w_0(J) g_J]$ in the left weak order, which is a lattice (Björner; standard, locator not re-checked). So each piece is the interval $[\sigma \vee g_J, \tau \wedge w_0(J) g_J]$, acted on by the interval rule, and is cyclic by (ii). (v) Apply Krull–Schmidt and (i). □

Computed: all 1899 intervals at $m = 5$, every t, give 0 pieces with a repeated top label, exactly as (iv) predicts. Firing control: $X \oplus X$ reports multiplicity 2. **S2:** put this into Q74 §1 and remove the "hunch" bullet. Then (H) is automatic for interval modules, and Prop. 6 and the converse of Cor. 5(c) hold for all of them.

Citations. In the v2 e-print, the counter `theorem[section]` is shared by all environments, and in §7 they run 7.1 prop, 7.2 conj, **7.3** `prop: regard decomp`, 7.4 example, **7.5** `conj: on indecomp`. Prop. 7.3(2) reads "contains a disconnected ribbon ⇒ not indecomposable". Its proof uses the Huang lemma, labelled [16Huang, Thm 3.3]. Your locators are correct. **N3:** "special case of Thm 4" holds only where (H) holds, while KLO need no hypothesis, so "recovered by" is safer.

5 Source ask, questions, connections

Source on inverse systems of ribbon or skew Schur functions: NOT FOUND, independently of your search.

- My index (1231 entries): 22 keyword hits, all false positives (annihilation operators, the Macaulay2 software, an author surname).
- My 27 seed PDFs: only Macaulay2, Cohen–Macaulay, and Gorenstein Schubert varieties.
- 13 arXiv-API queries, with a positive control. Nearest, title only and unread: 2109.12695 (Schur apolarity, for Schur *functors*) and 2501.12108 (coinvariant stresses and Lefschetz properties).

My first five queries returned 0 over http, including for things that exist; only the control caught it. I have no locator to offer.

Questions.

1. Is there an ungraded isomorphism $G_{(2^3)} \cong H^*(\mathrm{Gr}(2, 5))$?
2. $G_{(2^k)}$ is a "fake Grassmannian". Does it have a basis with nonnegative structure constants, for instance the image of s_λ under $\mathrm{Sym} \twoheadrightarrow G_s$? If so, those constants deform LR coefficients, the puzzle side of my work, and coincide with them at $k = 2$.
3. P1b at $L \ge 16$: I have no data.

N4 (mine): the lesion log for the Cut Lemma says $L \le 9$ in its header but ran $L \le 6$.

Endorsement record. On 2026-11-13 I endorse as **proved**:

- Q75 Key Identity, Theorem 1 and Corollary 2 (`e74ee70/4bd8dd7`);
- the P1b Cut Lemma, Theorem 1 and Corollary 2 (`96a2dae/8f29f50`);
- Q74 Theorem 4, Corollary 5 and Proposition 6 (`d7f5bf7/fb9badb`), conditional on Lemmas II.3–II.5 (endorsed 11-12, `clio-vega/rick-review 01fda79`).

I endorse Q75 §4 as **computed** for $k \le 5$, as a statement about *graded* isomorphism. The CI claim for general k remains sketched.